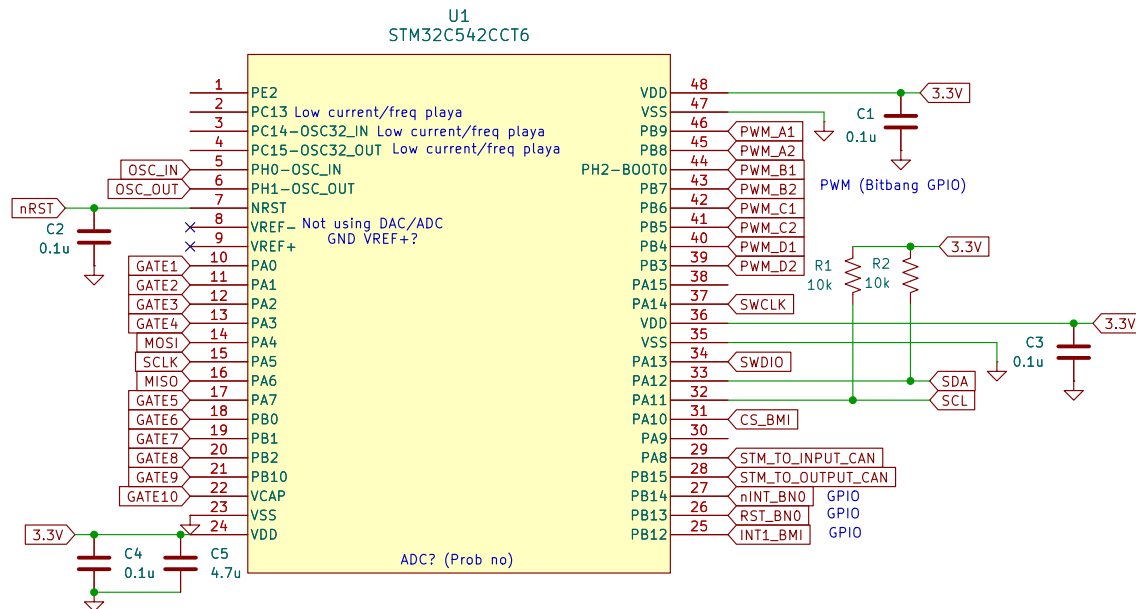


## MCU Connections



## Categorized Designs

### MOSFET\_driver

File: untitled.kicad\_sch

### Sensors\_Telemetry

File: Sensors\_Telemetry.kicad\_sch

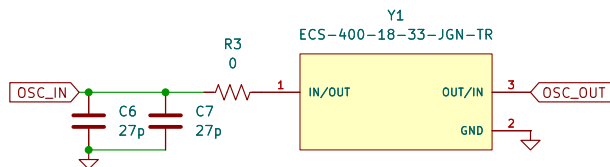
### Connector\_Signals

File: connector\_signals.kicad\_sch

### Voltage\_Conv-Protec-Monitoring

File: Voltage\_conv\_prot.kicad\_sch

## Crystal Oscillator



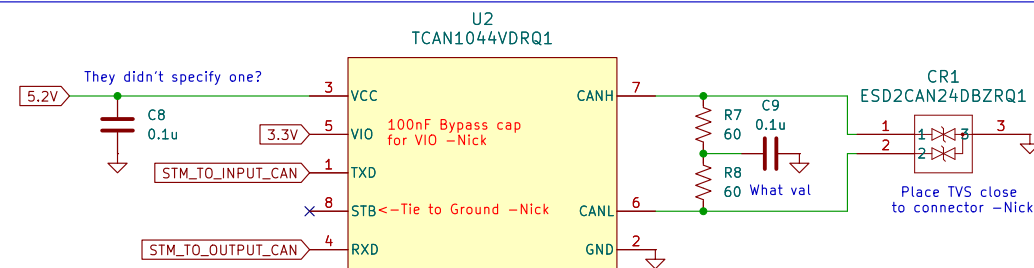
## MCU:

- add debug LED to STM GPIO
- break out a few GPIO STM pins to a pin header
- why do gpio bitbang for PWM? Cant we just use the STM's timer channels?
- add no connect (NC) to the unused STM pins

## General:

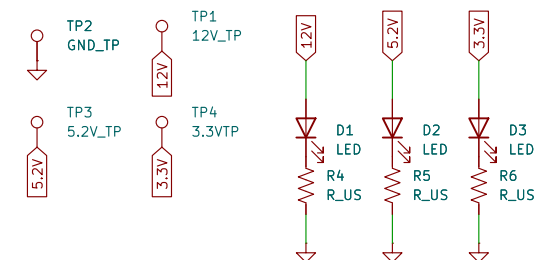
- testpoints and debugging:
  - break out CAN signals to a header for debugging
  - break out SPI

## CAN Transceiver



## Test Points, LEDs

## Resistor Values? - Nick



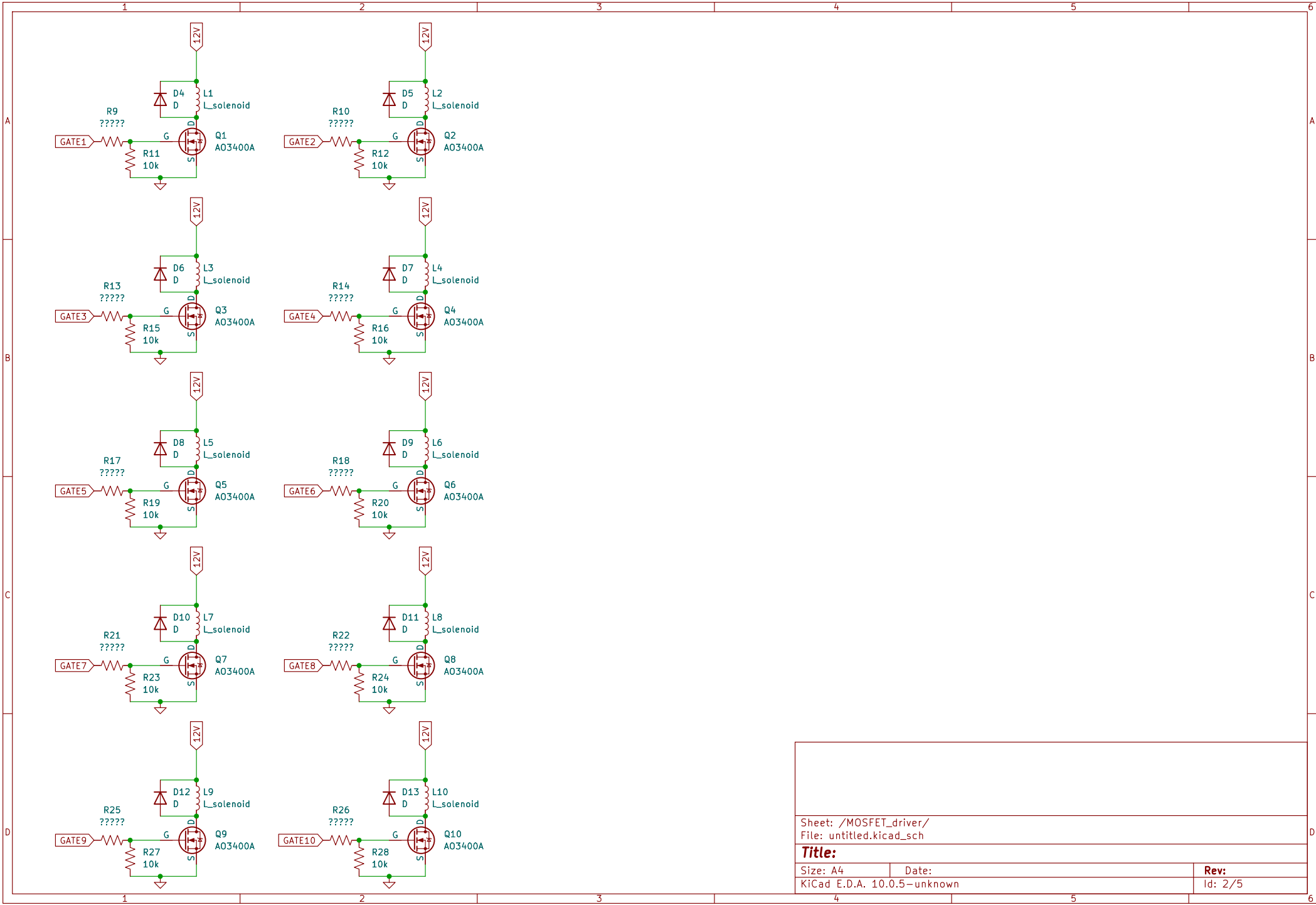
## Purdue ROV

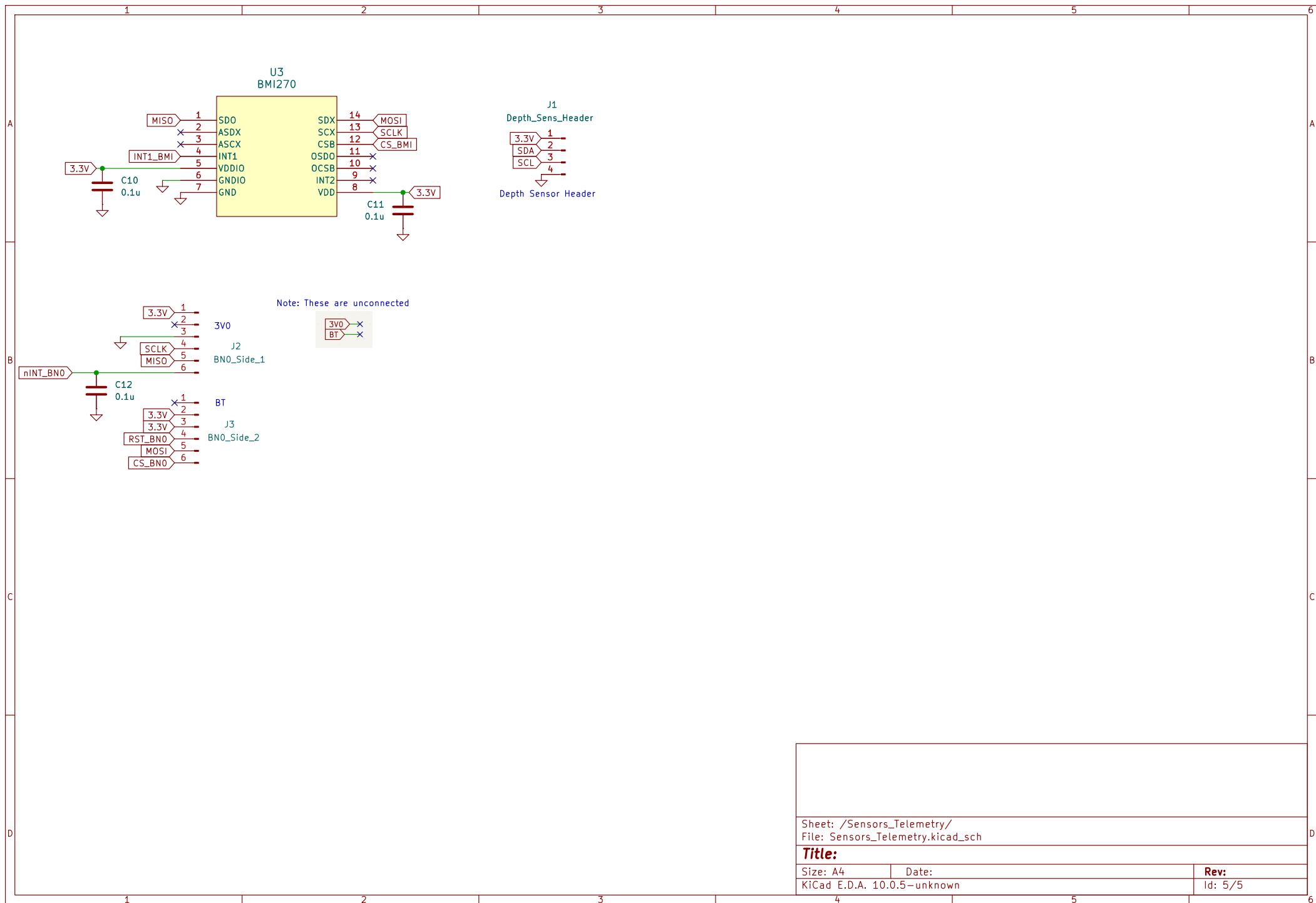
Sheet: /  
File: X19-Control-Board.kicad\_sch

## Title: Purdue ROV Subsystem Board

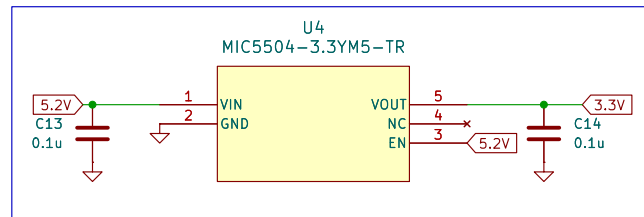
Size: A4  
KiCad E.D.A. 10.0.5-unknown

Date:  
Rev:  
Id: 1/5

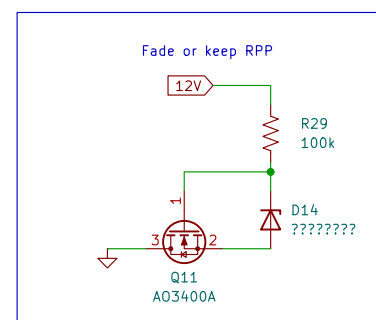




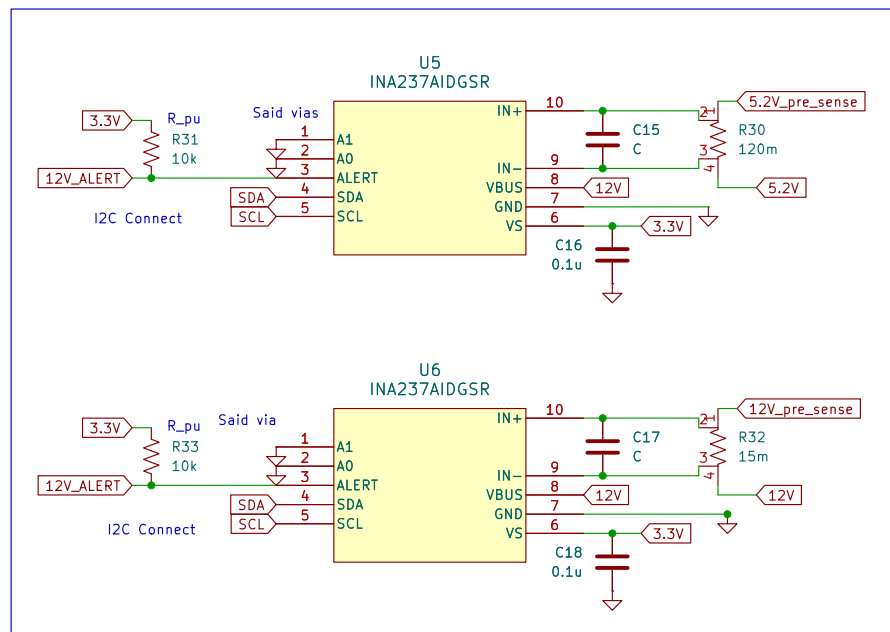
# LDO – Voltage Conversion



# Reverse Polarity Protection



# Power Monitoring 5.2V, 12V rails



Sheet: /Voltage Conv-Protec-Monitoring/  
File: Voltage\_conv\_prot.kicad\_sch

## Title:

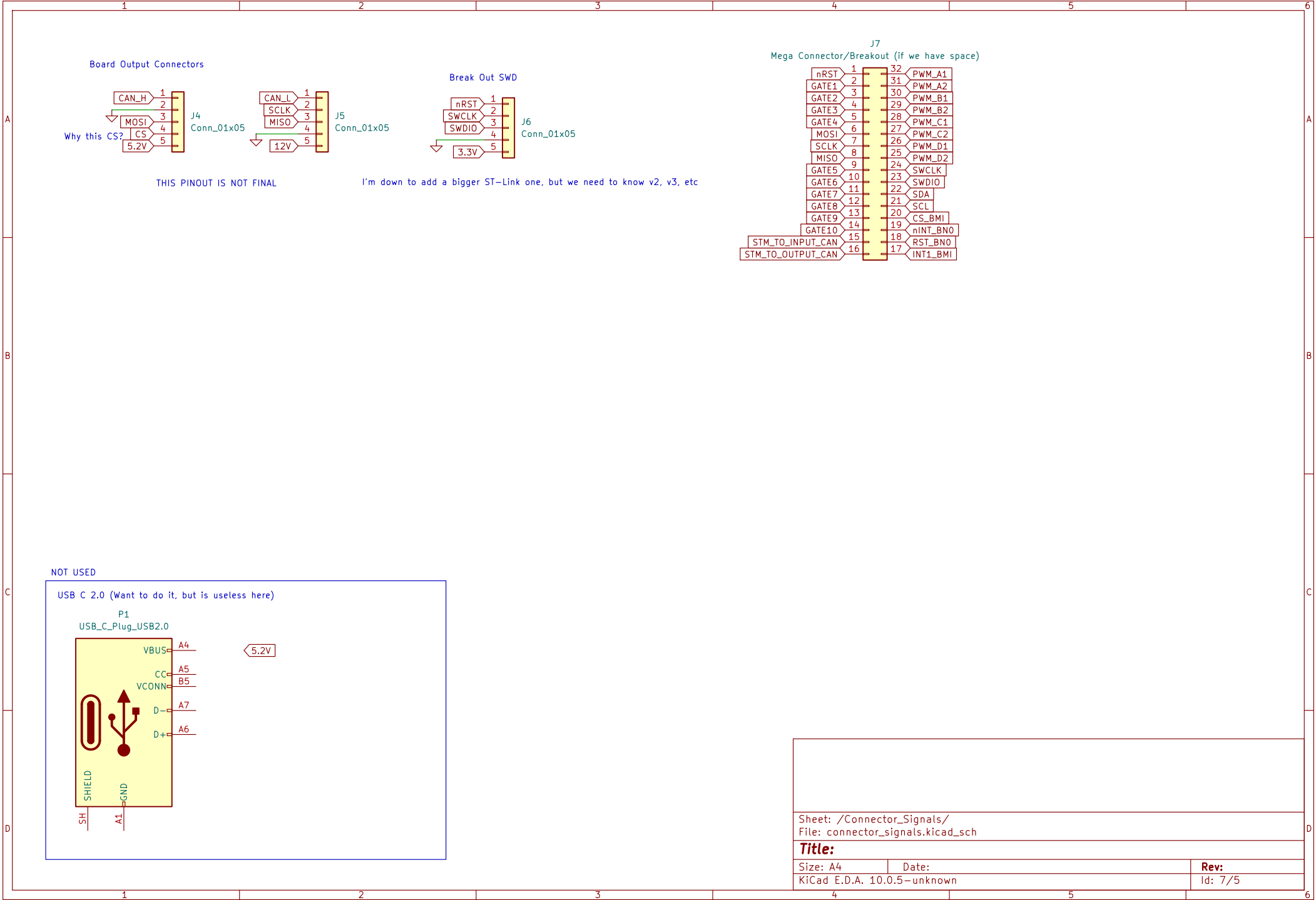
Size: A4

Date:

KiCad E.D.A. 10.0.5-unknown

Rev:

Id: 6/5



THIS PINOUT IS NOT FINAL

I'm down to add a bigger ST-Link one, but we need to know v2, v3, etc

Break Out SWD

nRST

1

2

SWCLK

3

SWDIO

4

3.3V

5

J6

Conn\_01x05

Mega Connector/Breakout (if we have space)

nRST

1

2

GATE1

3

GATE2

4

GATE3

5

GATE4

6

MOSI

7

SCLK

8

MISO

9

GATE5

10

GATE6

11

GATE7

12

GATE8

13

GATE9

14

GATE10

15

STM\_TO\_INPUT\_CAN

16

32

31

30

29

28

27

26

25

24

23

22

21

20

19

18

17

PWM\_A1

32

PWM\_A2

31

PWM\_B1

30

PWM\_B2

29

PWM\_C1

28

PWM\_C2

27

PWM\_D1

26

PWM\_D2

25

SWCLK

24

SWDIO

23

SDA

22

SCL

21

CS\_BMI

20

nINT\_BN0

19

RST\_BN0

18

INT1\_BMI

17

NOT USED

USB C 2.0 (Want to do it, but is useless here)

P1

USB\_C\_Plug\_USB2.0

VBUS

A4

CC

A5

VCONN

B5

D-

A7

D+

A6

SHIELD

SH

GND

A1

5.2V

Sheet: /Connector\_Signals/  
File: connector\_signals.kicad\_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.5-unknown

Rev:

Id: 7/5